

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, Java, TypeScript, C++, SQL

Tools and Frameworks: REST(Django, FastAPI), React, Next.js, PostgreSQL, PyTest, Selenium, AWS, Redis, Docker, Kubernetes, GitHub Actions(CI/CD), PyTorch, LangChain, CrewAI, Qdrant, Claude Code

Areas of Interest and Coursework: Distributed System Design, AI & LLMs, Software Engineering (SDLC)

INDUSTRY EXPERIENCE

Data AI Intern | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Developed backend services revolving around REST APIs on an AI-native GRC platform automating NIST 800-53 compliance workflows for ~1,200 controls for RFP bidders, cutting reviews from **days of manual work to minutes**.
- Implemented a fallback mechanism across 5 embedding models (OpenAI, Ollama) that re-runs **RAG** queries with insufficient control citation hits, lifting retrieval from **32/50 to 50/50** hand-reviewed compliance queries.
- Built the React interface for plan upload, gap-report review, and follow-up Q&A on unmet controls, along with exposing embedding and chat model selection options to users.
- Diagnosed an OOM bug where freshly pulled 4–5 GB LLM weights lingered in the OS page cache; fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built a React + Django AI assignment-review platform to flag rubric mistakes, cutting coach prep from **5 to 2 hours**.
- Architected a fully serverless AWS stack using an SQS-triggered Lambda AI worker, DynamoDB, and S3/CloudFront, sustaining 200+ submissions across 5 cohorts at ~\$3/month idle cost.

Full Stack Developer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Engineered an admin-configurable cost estimation engine, cutting quote turnaround from ~2 days to <1 min.
- Shipped an admin pricing console that cut price-change lead time from a **developer deploy** to an **instant edit**, letting non-technical staff run seasonal updates unassisted.
- Implemented idempotent Stripe webhooks, moving deposit collection online with **zero** duplicate charges.
- Developed and shipped the React SPA with client-side routing, role-aware protected routes and form state validated against the same schema as the API, serving customer and admin experiences from one bundle.
- Sustained **300 req/s** at <100 ms p95 latency on one EC2 instance via Redis caching and PostgreSQL indexing.

ACADEMIC EXPERIENCE

Graduate Assistant | UMBC - Computer Architecture & Software Engineering

Aug 2025 – June 2026

- Designed a C++ assignment on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

Teaching Assistant | UMBC - Data Structures

Jan 2025 – June 2025

- Developed 20+ unit tests for the hash table project, automating logic checking and reducing grading time by **80%**.
- Helped **50+** students design and debug C++ projects implementing core data structures and graded homework.

PROJECTS

When Pixels Talk Back | Watermarking EMR Data to analyse impact on CV models

- Embedded a 1024-bit blind fragile watermark into RONI regions of 2,482 COVID-19 lung CT scans, achieving near-lossless quality (SSIM **0.9999**) while preserving DenseNet-169 performance (F1 **0.952**).
- Benchmarked 3 watermarking algorithms against 3 Computer Vision models, showing robust watermarks cut positive-case detection to **48%** and skew errors toward false negatives.

Nothing Lost in Translation | Lexically constrained Sequence Generation

- Built a Grid Beam Search decoder guaranteeing positive/negative lexical constraint satisfaction and raising BLEU by **15 points** (25.45 → 40.59) for English-to-Russian NMT.